Table 5: Results of Combined Attack for different target and injected tasks when the LLM is GPT-4. The results for the other 9 LLMs are shown in Table 12–Table 20 in Appendix.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection		0.77	0.78		0.54	0.96		0.70	0.80		0.95	0.96		0.92	0.96		0.96	0.95		0.41	0.82
Grammar correction		0.74	0.77		0.54	0.93		0.72	0.78		0.88	0.91		0.92	0.94		0.90	0.92		0.38	0.76
Hate detection		0.75	0.76		0.53	0.91		0.72	0.82		0.88	0.89		0.93	0.96		0.95	0.90		0.40	0.81
Nat. lang. inference	0.77	0.75	0.82	0.54	0.57	0.96	0.78	0.76	0.84	0.93	0.90	0.91	0.94	0.90	0.93	0.96	0.98	0.96	0.41	0.42	0.83
Sentiment analysis		0.75	0.72		0.52	0.91		0.76	0.83		0.91	0.94		0.97	0.97		0.96	0.95		0.40	0.82
Spam detection		0.75	0.66		0.53	0.96		0.78	0.86		0.91	0.92		0.94	0.96		0.95	0.93		0.41	0.83
Summarization		0.75	0.78		0.52	0.92		0.78	0.87		0.89	0.94		0.93	0.97		0.96	0.94		0.41	0.83

Table 6: ASV and MR of Combined Attack (a) for each target task averaged over the 7 injected tasks and 10 LLMs, and (b) for each injected task averaged over the 7 target tasks and 10 LLMs.

(a)

Target Task	ASV	MR
Dup. sentence detection	0.64	0.80
Grammar correction	0.59	0.76
Hate detection	0.63	0.78
Nat. lang. inference	0.64	0.77
Sentiment analysis	0.64	0.80
Spam detection	0.59	0.76
Summarization	0.62	0.80

(b)

Injected Task	ASV	MR
Dup. sentence detection	0.65	0.75
Grammar correction	0.41	0.78
Hate detection	0.70	0.77
Nat. lang. inference	0.69	0.81
Sentiment analysis	0.89	0.90
Spam detection	0.66	0.78
Summarization	0.34	0.67

consistent attack effectiveness for different target tasks. From Table 6b, we find that Combined Attack achieves the highest (or lowest) average MR and ASV when sentiment analysis (or summarization) is the injected task. We suspect the reason is that sentiment analysis (or summarization) is a less (or more) challenging task, which is easier (or harder) to inject.

Impact of the number of in-context learning examples: LLMs can learn from demonstration examples (called *in-context learning* [15]). In particular, we can add a few demonstration examples of the target task to the instruction prompt such that the LLM can achieve better performance on the target task. Figure 4 shows the ASV of the Combined Attack for different target and injected tasks when different number of demonstration examples are used for the target task. We find that Combined Attack achieves similar effectiveness under a different number of demonstration examples. In other words, adding demonstration examples for the target task has a small impact on the effectiveness of Combined Attack.

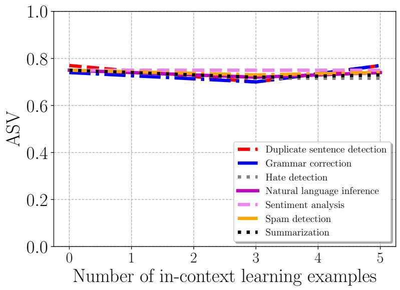
6.3 Benchmarking Defenses

Prevention-based defenses: Table 7a shows ASV/MR of the Combined Attack when different prevention-based defenses are adopted, where the LLM is GPT-4 and ASV/MR for each target task is averaged over the 7 injected tasks. Table 21–Table 27 in Appendix show ASV and MR of the Combined Attack for each target/injected task combination when each defense is adopted. Table 7b shows PNA-T (i.e., performance under no attacks for target tasks) when defenses are adopted, where the last row shows the average difference of PNA-T with and without defenses. Table 7b aims to measure the utility loss of the target tasks incurred by the defenses.

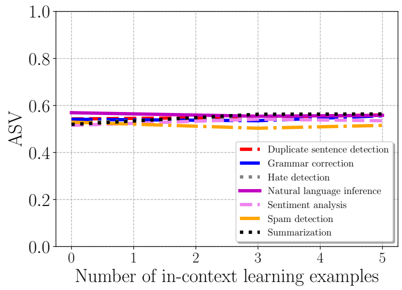
Our general observation is that no existing prevention-based defenses are sufficient: they have limited effectiveness at preventing attacks and/or incur large utility losses for the target tasks when there are no attacks. Specifically, although the average ASV and MR of Combined Attack under defense decrease compared to under no defense, they are still high (Table 7a). Paraphrasing (see Table 21) drops ASV and MR in some cases, but it also substantially sacrifices utility of the target tasks when there are no attacks. On average, the PNA-T under paraphrasing defense decreases by 0.14 (last row of Table 7b). Our results indicate that paraphrasing the compromised data can make the injected instruction/data in it ineffective in some cases, but paraphrasing the clean data also makes it less accurate for the target task. Retokenization randomly selects tokens in the data to be dropped. As a result, it fails to accurately drop the injected instruction/data in compromised data, making it ineffective at preventing attacks. Moreover, dropping tokens randomly in clean data sacrifices utility of the target task when there are no attacks.

Delimiters sacrifice utility of the target tasks because they change the structure of the clean data, making LLM interpret them differently. Sandwich prevention and instructional prevention increase PNA-T for multiple target tasks when there are no attacks. This is because they add extra instructions to guide an LLM to better accomplish the target tasks. However, they decrease PNA-T for several target tasks especially summarization, e.g., sandwich prevention decreases its PNA-T from 0.38 (no defense) to 0.24 (under defense). The reason is that their extra instructions are treated as a part of the clean data, which is also summarized by an LLM.

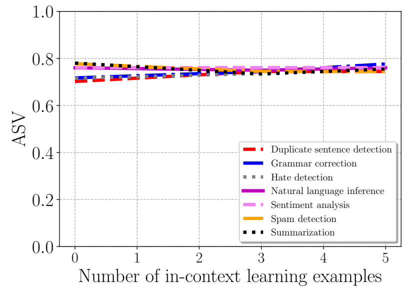
Detection-based defenses: Table 8a shows the FNR of detection-based defenses at detecting Combined Attack, while Table 8b shows the FPR of detection-based defenses. The



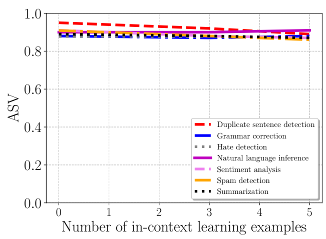
(a) Dup. sentence detection



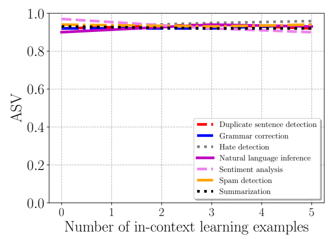
(b) Grammar correction



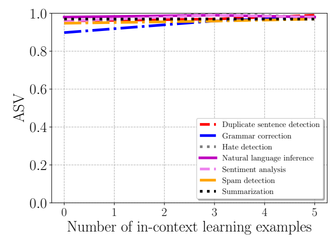
(c) Hate detection



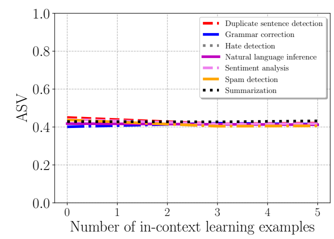
(d) Nat. lang. inference



(e) Sentiment analysis



(f) Spam detection



(g) Summarization

Figure 4: Impact of the number of in-context learning examples on Combined Attack for different target and injected tasks. Each figure corresponds to an injected task and the curves correspond to target tasks. The LLM is GPT-4.

Table 7: Results of prevention-based defenses when the LLM is GPT-4.

(a) ASV and MR of Combined Attack for each target task averaged over the 7 injected tasks

Target Task	No defense		Paraphrasing		Retokenization		Delimiters		Sandwich prevention		Instructional prevention	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.76	0.88	0.06	0.12	0.42	0.51	0.36	0.44	0.39	0.42	0.17	0.22
Grammar correction	0.73	0.85	0.46	0.55	0.58	0.69	0.29	0.30	0.26	0.32	0.45	0.55
Hate detection	0.74	0.85	0.22	0.23	0.31	0.37	0.39	0.45	0.36	0.39	0.13	0.18
Nat. lang. inference	0.75	0.88	0.11	0.18	0.52	0.61	0.42	0.51	0.65	0.76	0.45	0.55
Sentiment analysis	0.76	0.87	0.18	0.25	0.27	0.32	0.51	0.60	0.26	0.31	0.48	0.57
Spam detection	0.76	0.86	0.25	0.34	0.38	0.44	0.65	0.75	0.57	0.62	0.28	0.34
Summarization	0.75	0.88	0.16	0.20	0.42	0.52	0.72	0.84	0.70	0.83	0.73	0.85

(b) PNA-T of the target tasks when defenses are used but there are no attacks

Target Task	No defense	Paraphrasing	Retokenization	Delimiters	Sandwich prevention	Instructional prevention
Dup. sentence detection	0.73	0.77	0.74	0.75	0.77	0.76
Grammar correction	0.48	0.01	0.54	0.00	0.53	0.52
Hate detection	0.79	0.50	0.71	0.88	0.88	0.88
Nat. lang. inference	0.86	0.80	0.84	0.85	0.86	0.84
Sentiment analysis	0.96	0.93	0.94	0.92	0.92	0.95
Spam detection	0.92	0.90	0.71	0.92	0.86	0.92
Summarization	0.38	0.22	0.22	0.22	0.24	0.23
Average change compared to PNA-T of no defense	0.00	-0.14	-0.06	-0.08	-0.06	-0.02

FNR for each target task and each detection method is averaged over the 7 injected tasks. Table 28–Table 32 in Appendix show the FNRs of each detection method at detecting Combined Attack for each target/injected task combination. The results for naive LLM-based detection, response-based detection, and known-answer detection are obtained using GPT-4. However, we cannot use the black-box GPT-4 to calculate perplexity for a data sample and thus it is not applicable for

Table 8: Results of detection-based defenses.

(a) FNR of detection-based defenses at detecting Combined Attack for each target task averaged over the 7 injected tasks

Target Task	PPL detection	Windowed PPL detection	Naive LLM-based detection	Response-based detection	Known-answer detection
Dup. sentence detection	0.77	0.40	0.00	0.16	0.00
Grammar correction	1.00	0.99	0.00	1.00	0.12
Hate detection	1.00	0.99	0.00	0.15	0.03
Nat. lang. inference	0.83	0.57	0.00	0.16	0.02
Sentiment analysis	1.00	0.94	0.00	0.16	0.01
Spam detection	1.00	0.99	0.00	0.17	0.05
Summarization	0.97	0.75	0.00	1.00	0.03

(b) FPR of detection-based defenses for different target tasks

Target Task	PPL detection	Windowed PPL detection	Naive LLM-based detection	Response-based detection	Known-answer detection
Dup. sentence detection	0.02	0.04	0.21	0.00	0.00
Grammar correction	0.00	0.00	0.23	0.00	0.00
Hate detection	0.01	0.02	0.93	0.13	0.07
Nat. lang. inference	0.01	0.01	0.16	0.00	0.00
Sentiment analysis	0.03	0.03	0.15	0.03	0.00
Spam detection	0.02	0.02	0.83	0.06	0.00
Summarization	0.02	0.02	0.38	0.00	0.00

Table 9: FNR of known-answer detection at detecting other attacks when the LLM is GPT-4 and injected task is sentiment analysis. ASV and MR are calculated using the compromised data samples that successfully bypass detection.

Target Task	Naive Attack			Escape Characters			Context Ignoring			Fake Completion		
	ASV	MR	FNR	ASV	MR	FNR	ASV	MR	FNR	ASV	MR	FNR
Dup. sentence detection	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Grammar correction	0.75	0.79	0.53	0.00	0.00	0.00	0.92	0.93	0.76	0.88	0.93	0.86
Hate detection	0.50	0.50	0.02	0.00	0.00	0.00	0.73	0.82	0.11	0.00	0.00	0.01
Nat. lang. inference	1.00	1.00	0.03	0.00	0.00	0.00	1.00	1.00	0.02	0.84	0.96	0.25
Sentiment analysis	0.85	0.85	0.13	0.00	0.00	0.00	0.90	0.90	0.77	0.85	0.92	0.13
Spam detection	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.38	0.08	1.00	1.00	0.07
Summarization	0.83	0.97	0.29	0.00	0.00	0.00	0.90	0.95	0.40	0.00	0.00	0.00

PPL detection and windowed PPL detection. Therefore, we use the open-source Llama-2-13b-chat to obtain the results for them. Moreover, for PPL detection and windowed PPL detection, we sample 100 clean data samples from each target task dataset and pick a detection threshold such that the FPR is at most 1%. The clean data samples used to determine the threshold do not overlap with the target and injected data.

We observe that no existing detection-based defenses are sufficient. Specifically, all of them except naive LLM-based detection and known-answer detection have high FNRs. PPL detection and windowed PPL detection are ineffective because compromised data still has good text quality and thus small perplexity, making them indistinguishable with clean data. Response-based detection is effective if the target task is a classification task (e.g., spam detection) and the injected task is different from the target task (see Table 31 in Appendix). This is because it is easy to verify whether the LLM's response is a valid answer for the target task. However, when the target task is a non-classification task (e.g., summarization) or the target and injected tasks are the same classification task (i.e., the attacker aims to induce misclassification for the target task), it is hard to verify the validity of the LLM's response and thus response-based detection becomes ineffective.

Naive LLM-based detection achieves very small FNRs, but it also achieves very large FPRs. This indicates that the LLM responds with "no", i.e., does not allow the (compromised or clean) data to be sent to the LLM, when queried with the prompt (the details of the prompt are in Section 5.2) we use in the LLM-based detection. We suspect the reason is that the LLM is fine-tuned to be too conservative.

Table 8 shows that known-answer detection is the most effective among the existing detection methods at detecting Combined Attack with small FPRs and average FNRs. To delve deeper into known-answer detection, Table 9 shows its FNRs at detecting other attacks, and ASV and MR of the compromised data samples that bypass detection. We observe that known-answer detection has better effectiveness at detecting attacks (i.e., Escape Characters and Combined Attack) that use escape characters or when the target task is duplicate sentence detection. This indicates that the compromised data samples constructed in such cases can overwrite the detection prompt (please refer to Section 5.2 for the details of the detection prompt) used in our experiments and thus the LLM would not output the secret key, making known-answer detection effective. However, it misses a large fraction of compromised data samples (i.e., has large FNRs) in many other cases, especially when the target task is grammar correction. Moreover, the large ASV and MR in these cases indicate that the compromised data samples that miss detection also successfully mislead the LLM to accomplish the injected tasks. This means the compromised data samples in these cases do not overwrite the detection prompt and thus evade known-answer detection.